

## Day 4 – Patterns and looping

1. Square pattern
2. Right angle triangle star pattern
3. Inverted right angle triangle star pattern
4. Left angle triangle pattern
5. Inverted left angle triangle pattern
6. Pyramid pattern
7. Inverted pyramid pattern
8. Diamond pattern
9. Print Prime Numbers in a Range Using for Loop
10. Generate Fibonacci Series up to N Terms
11. Prime or not

## Day 4 – Patterns and looping

### 1. Square pattern

```
n = int(input("Enter size: "))
for i in range(1,n+1):
    for j in range(1,n):
        print("*", end=" ")
    print()
```

### 2.Right angle star pattern

```
n = int(input("Enter rows: "))
for i in range(1, n + 1):
    for j in range(i):
        print("*", end=" ")
    print()
```

### 3.Inverted Right triangle

```
n = int(input("Enter rows: "))
for i in range(n+1, 1, -1):
    for j in range(i):
        print("*", end=" ")
    print()
```

## Day 4 – Patterns and looping

### 4. Left angle triangle

```
n = int(input("Enter rows: "))  
for i in range(1, n + 1):  
    print(" " * (n - i), end="")  
    print("*" * i)
```

### 6.Pyramid

```
n = int(input("Enter rows: "))  
for i in range(1, n + 1):  
    print(" " * (n - i), end="")  
    print("*" * (2 * i - 1))
```

### 7.Inverted pyramid

```
n = int(input("Enter rows: "))  
for i in range(n+1, 1, -1):  
    print(" " * (n - i), end="")  
    print("*" * (2 * i - 1))
```

## Day 4 – Patterns and looping

### 8. Diamond pattern

```
n = int(input("Enter rows: "))
for i in range(1, n + 1):
    print(" " * (n - i), end="")
    print("*" * (2 * i - 1))
for i in range(n, 1, -1):
    print(" " * (n - i), end="")
    print("*" * (2 * i - 1))
```

### 9. Print Prime Numbers in a Range Using for Loop

```
start = int(input("Enter start: "))
end = int(input("Enter end: "))
for num in range(start, end + 1):
    if num > 1:
        prime = True
        for i in range(2, num):
            if num % i == 0:
                prime = False
                break
    if prime:
        print(num)
```

## 10. Generate Fibonacci Series up to N Terms

```
n = int(input("Enter number of terms: "))
a = 0
b = 1
for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

## 11.Prime or nor

```
num = int(input("Enter a number: "))
if num <= 1:
    print("Not a Prime Number")
else:
    prime = True
    for i in range(2, num):
        if num % i == 0:
            prime = False
            break
    if prime:
        print("Prime Number")
    else:
        print("Not a Prime Number")
```